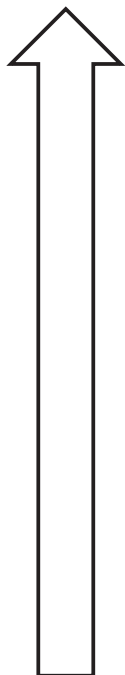




11.7.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement which best describes your current understanding of solving real-world problems with rational numbers.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will make sure that I get help.

I do not understand any of this yet. There are things I could do to be a better learner.



Unit 11, Lesson 8: Evaluating Numeric Expressions – Part 1



Benchmarks

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

- ☐ I can evaluate numeric expressions with up to four operations.



11.8.1 Warm-Up: Math Symbols

In the 18th century, Robert Recorde decided to use a symbol for the idea of “is equal to” so he would not have to write the words over and over. As more mathematicians also began to use the symbol, the “=” became accepted as the standard symbol for equality. Math is full of symbols that shorten lengthy words and calculations. But just like symbols you use when communicating with others, like “LOL” in texting, once you learn their meaning, understanding the mathematics is not so challenging.

1. List math symbols that you are familiar with and what they represent.
2. List other math symbols you have seen before, but you are unfamiliar with.



11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations.

Step 1: Perform operations within grouping symbols.

Step 2: Multiply and divide in order from left to right ($\rightarrow$).

Step 3: Add and subtract in order from left to right ($\rightarrow$).

Numerical expressions can also include exponents.

1. The expression $4 \times (-2)^3 - 8 \div (-3 + 1)$ can be rewritten by expanding the exponential expression as $4 \times (-2 \cdot -2 \cdot -2) - 8 \div (-3 + 1)$.
 - a. According to the order of operations, which expressions should be evaluated first?
 - b. When $(-2)^3$ is rewritten using multiplication, the resulting expression is in parentheses. Complete the statement.

According to the order of operations, $(-2)^3$ should

be evaluated

after
before

 multiplying by 4.

- c. Work with your partner to number each operation so that the order of operations is in the correct order.

- ☐ Perform operations inside grouping symbols.
- ☐ Perform multiplication and division in order from left to right. ($\rightarrow$)
- ☐ Evaluate expressions with exponents.
- ☐ Perform addition and subtraction in order from left to right. ($\rightarrow$)

2. Use order of operations to evaluate one of the expressions shown in the table below while your partner evaluates the other expression.

Expression A	Expression B
$(6 + 0.5)^2 \div 0.25$	$(6^2 + 0.5^2) \div 0.25$

3. How are the problems similar?

4. How are the problems different?

5. Complete the statements about the expressions.

The two expressions are ☐ equivalent
☐ not equivalent

because they ☐ have the same value.
☐ do not have the same value.

For Expression A, first

- ☐ add inside the parentheses, then apply the exponent to the sum.
- ☐ apply the exponent to each addend, then add to find the sum.

For Expression B, first

- ☐ add inside the parentheses, then apply the exponent to the sum.
- ☐ apply the exponent to each addend, then add to find the sum.

For both expressions, addition was done before division

because ☐ the addition is to the left of the division.
☐ the addition is inside the grouping symbols.

6. Anna evaluated Expression A and Expression B. Her work is shown.

Expression A	Expression B
$ -10 \div 2 + 10^2$	$-10 \div 2 + 10^2$
$10 \div 2 + 10^2$	$-10 \div 2 + 100$
$10 \div 2 + 100$	$-5 + 100$
$5 + 100$	95
105	

Anna's partner gave her a 👍 on both of her answers. Why did Anna arrive at two different values when evaluating these similar looking expressions?



Absolute value notation should be treated the same as grouping symbols when following the order of operations.

7. Zion says since multiplication is commutative and the order does not matter, he thinks he can evaluate the expression $[0.05 \times (3 \times 2)^4]^2$ by either multiplying 0.05×3 first or by multiplying 3×2 first.

a. Discuss with your partner if Zion is correct when he stated that both would work.

b. Evaluate the expression $[0.05 \times (3 \times 2)^4]^2$.



When using the order of operations to evaluate an expression with nested grouping symbols, apply the order of operations inside the inner grouping symbols first.



11.8.3 Collaboration: Determining Equivalent Expressions

Answer the questions with your partner.

Tanisha says that when she evaluated an expression, the result was 7.25.

- Use mental math to estimate each value. Then, circle which of the following expressions could have been Tanisha's expression.

$$(6 - 0.5)^2 \times 5$$

$$6 - 0.5^2 \times 5$$

$$6 + (-0.5)^2 \times 5$$

$$5 \times 6 - 0.5^2$$

- Find the values of the expressions to verify your choice.

Expression	Value
$(6 - 0.5)^2 \times 5$	
$6 - 0.5^2 \times 5$	
$6 + (-0.5)^2 \times 5$	
$5 \times 6 - 0.5^2$	



11.8.4 Exploration: Grouping Symbols

- Without working the problems, discuss with your partner which expression in the table you expect to have the largest value.

Expression A	Expression B
$10 - (0.75 \times 12) + 4^2$	$(10 - 0.75) \times (12 + 4^2)$
Expression C	Expression D
$10 - 0.75 \times 12 + 4^2$	$10 - 0.75 \times (12 + 4)^2$

- Put an asterisk (*) next to the expression you expect will have the largest value. Write your explanation here.
- Evaluate each of the expressions in the table with your partner.

4. With another set of partners, share if your guess was or was not correct. Discuss how the order of operations affected the value of your guesses. Summarize your discussion.

5. For the two expressions in the table that resulted in the same value, explain why they have the same value.



11.8.5 Your Turn!

1. Evaluate each expression without using a calculator.

a. $[(10 - 15) \times 2]^3 + 50$

b. $(2^4 - 3 \times 6)^2$

c. $|18 + (-2) \times 14|^5$

d. $[0.8 \times (1.2 + 2.8)^2]^2$

e. $[(0.8 \times 10)^2 + 4.5] \div 0.5$

f. $(10 \times 0.4)^2 + (28.4 \div (-4))$



11.8.6 Wrap-Up: Reflection

1. Complete each statement based on your learning today.

- A barrier to my learning was ...
- I will try to overcome this by ...
- Today I am still unsure about ...
- To fill in this gap I intend to ...



Unit 11, Lesson 9: Evaluating Numeric Expressions – Part 2



Benchmarks

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

- ☐ I can evaluate numeric expressions with up to six operations.